



6669 - A massive protocluster at z=4.3 selected by the South Pole Telescope

Cycle: 3, Proposal Category: GO

INVESTIGATORS

<i>Name</i>	<i>Institution</i>
Dr. Scott C. Chapman (PI) (CSA Member)	University of British Columbia
Prof. Joaquin Vieira (CoI) (CoPI) (US Admin CoI)	University of Illinois at Urbana - Champaign

OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
SPT2349-56				
	1	MIRI Imaging	MIRI Imaging	(1) SPT2349-56-IMAGING
	2	NIRCAM Imaging	NIRCam Imaging	(1) SPT2349-56-IMAGING
	3	MIRI Imaging	MIRI Imaging	(1) SPT2349-56-IMAGING
	4	MIRI Imaging	MIRI Imaging	(1) SPT2349-56-IMAGING

ABSTRACT

accompanying APT for JWST component of joint program with HST

Note that charged time in this new APT has shifted from the original 4.74hrs, to 4.83hrs. The science time and exposure parameters remain identical to that proposed in the joing HST+JWST submission.

OBSERVING DESCRIPTION

Along with the 3 filter HST observations, we request MIRI imaging and NIRCAM imaging.

The total JWST request is 2.42 hours on source, 4.83hr charged, with depths leveraged from SED analysis (Fig 4) to detect all SMGs (MIRI) and all UV-selected galaxies in exist- ing imaging (NIRCam). For the dedicated NIRCam setting, we select the F444W band in the long wavelength channel,

JWST Proposal 6669 (Created: Monday, August 19, 2024 at 11:00:28 AM Eastern Standard Time) - Overview

with F200W in the short wavelength channel (3270s) to achieve improved depth and resolution compared with HST. The HST F160W and NIRCам F200W/F444W bands allow a measurement of the SED slope across the Balmer break, an age-sensitive feature that we will use to constrain the presence and age of any older stellar population. We select the FULLBOX dither pattern because it covers targets and candidate members of the proto-cluster, and choose the readout pattern and groups/integration using the JDox-recommended patterns for maximal depth in a given exposure time.

For MIRI, we will observe in the F1000W (1443s) and F1800W (3996s) filters, which with NIRCам F444W cover the 1.6m stellar bump to allow robust masses, especially for the highly-obscured sources. We use a 4-point dither pattern and FAST readout mode with 100 groups per integration for all filters, sufficient to reach high S/N for all sources given the steeply rising SEDs towards longer mid-IR wavelengths. MIRI/F1000W will cover the Pa- line and allow us to better constrain the SFR of objects in the wider field. MIRI/F1800W will cover the PAH 3.3m line and allow us to better constrain the SFR of objects in the wider field and the central BCG. The PAH 3.3 m is known to be a key indicator of star formation in local LIRGs and ULIRGs [14]. Low equivalent width of this PAH feature (or lack of a PAH detection) is also expected to be an indicator of AGN presence, informative for dust enshrouded star forming galaxies or AGN having visual band extinction 10-30 [14].

Proposal 6669 - Targets - A massive protocluster at $z=4.3$ selected by the South Pole Telescope

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	SPT2349-56-IMAGING	RA: 23 49 42.8400 (357.4285000d) Dec: -56 38 25.80 (-56.64050d) Equinox: J2000		
	Comments: Category=Galaxy Description=[High-redshift galaxies] Extended=YES				

Proposal 6669 - Observation 1 - A massive protocluster at z=4.3 selected by the South Pole Telescope

Observation	Proposal 6669, Observation 1: MIRI Imaging										Mon Aug 19 16:00:28 GMT 2024
	Diagnostic Status: Warning										
Diagnostics	Observing Template: MIRI Imaging										
	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(1)	SPT2349-56-IMAGING	RA: 23 49 42.8400 (357.4285000d) Dec: -56 38 25.80 (-56.64050d) Equinox: J2000								
Template	Comments: Category=Galaxy Description=[High-redshift galaxies] Extended=YES										
	Subarray										
Dithers	FULL										
	#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size	
Spectral Elements	1	CYCLING	1	4						DEFAULT	
	#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
Special Requirements	1	F1800W	FASTR1	90	4	1	Dither 1	4	16	4029.358	
	Offset 6.1186 arcsec, -33.33 arcsec										

Proposal 6669 - Observation 2 - A massive protocluster at z=4.3 selected by the South Pole Telescope

Observation	Proposal 6669, Observation 2: NIRCAM Imaging									Mon Aug 19 16:00:28 GMT 2024
	Diagnostic Status: Warning									
Observing Template: NIRCcam Imaging										
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.									
	(Visit 2:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.									
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous	
	(1)	SPT2349-56-IMAGING	RA: 23 49 42.8400 (357.4285000d) Dec: -56 38 25.80 (-56.64050d) Equinox: J2000							
Comments: Category=Galaxy Description=[High-redshift galaxies] Extended=YES										
Template	Module		Subarray				Target Placement			
	ALL		FULL				Module Gap			
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions
	1	FULLBOX		4		STANDARD				2
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID
	1	F200W	F444W	MEDIUM8	4	1	8	8	3263.978	
Special Requirements	Group Visits within 53.0 Days Visits Same PA Offset 100.0 arcsec, -20.0 arcsec									

Proposal 6669 - Observation 3 - A massive protocluster at z=4.3 selected by the South Pole Telescope

Observation	Proposal 6669, Observation 3: MIRI Imaging										Mon Aug 19 16:00:28 GMT 2024
	Diagnostic Status: Warning										
Diagnostics	Observing Template: MIRI Imaging										
	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(1)	SPT2349-56-IMAGING	RA: 23 49 42.8400 (357.4285000d) Dec: -56 38 25.80 (-56.64050d) Equinox: J2000								
Template	Comments: Category=Galaxy Description=[High-redshift galaxies] Extended=YES										
	Subarray										
Dithers	FULL										
	#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size	
	1	CYCLING	1	4						DEFAULT	
Spectral Elements	#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	F1000W	FASTR1	65	2	1	Dither 1	4	8	1454.121	
Special Requirements	Offset 6.1186 arcsec, -33.33 arcsec										

Proposal 6669 - Observation 4 - A massive protocluster at z=4.3 selected by the South Pole Telescope

Observation	Proposal 6669, Observation 4: MIRI Imaging										Mon Aug 19 16:00:28 GMT 2024
	Diagnostic Status: Warning										
	Observing Template: MIRI Imaging										
Diagnostics	(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(1)	SPT2349-56-IMAGING	RA: 23 49 42.8400 (357.4285000d)								
			Dec: -56 38 25.80 (-56.64050d)								
			Equinox: J2000								
Template	Comments: Category=Galaxy Description=[High-redshift galaxies] Extended=YES										
	Subarray										
	FULL										
Dithers	#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size	
	1	CYCLING	1	4						DEFAULT	
Spectral Elements	#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	F1800W	FASTR1	90	4	1	Dither 1	4	16	4029.358	
Special Requirements	Offset 6.1186 arcsec, -33.33 arcsec										